

Air Coolers

Brand	Cooler Master	Cooler Master	Deepcool	Cooler Master	Cooler Master	Deepcool	
Model	Hyper TX3 Evo	Hyper T4	GAMMAXX 400	Hyper 212 Evo	Hyper 212X	FROSTWIN	
Price (₹)	1,500	1,800	2,125	2,300	2,300	2,900	
Test Scores							
Features and Accessories (Out of 25)	14.31	14.25	15.5	14.25	14.25	12.25	
Performance (Out of 75)	54.82	61.92	62.25	63.3	63.43	64.53	
Grand Total (Out of Hundred)	69.13	76.17	77.75	77.55	77.68	76.78	
Specifications							
Intel CPU Socket LGA (775/1150/1155/1156/1366/2011)	(Y/Y/Y/Y/Y/N)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	
AMD CPU Socket (AM2/AM2+/AM3/AM3+/FM1/FM2)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	(Y/Y/Y/Y/Y/Y)	
Heatsink Dimensions (in mm)	90x51x136	128x44x152.3	135X76X159	116x51x159	116x51x158	121x121x151	
Heatsink(Single tower/Twin tower)	Single tower	Single Tower	Single Tower	Single Tower	Single Tower	Twin Tower	
Fan (Number/diameter in mm)	1/92mm	1/120mm	1/120mm	1/120mm	1/120mm	2/120mm	
Fan Speed (in RPM)	2800	1800	1500	2000	2000	1600	
Fan Bearing Type	Sleeve	Rifle	Hydro	Sleeve	4th Gen	Hydro	
Number of heatpipes	3	4	3	4	4	4	
Heatsink Weight (in gm)	306	448	709	465	492	712	
Features							
Cold Plate Material (Cu/Al/Cu+Al/Ni plated Cu) (Y/N)	(N/N/Y/N)	(N/N/Y/N)	(N/N/Y/N)	(N/N/Y/N)	(N/N/Y/N)	(N/N/Y/N)	
Vapor Chamber/PWM/Sheathed Cable	N/Y/N	N/Y/N	N/Y/N	N/Y/N	N/Y/N	N/Y/N	
Heat pipe contact type(direct/indirect)	Direct	Direct	Direct	Direct	Direct	Direct	
Accessories							
Screws/S Driver/T Paste/ Fan mounts	Y/N/Y/Y	Y/N/Y/Y	Y/N/Y/Y	Y/N/Y/Y	Y/N/Y/Y	Y/N/Y/N	
Bundled (Fan splitter/Molex Converter/ Low Noise Adapter) (Y/N)	(N/N/N)	(N/N/N)	(N/N/N)	(N/N/N)	(N/N/N)	(N/N/N)	
Performance							
Idle performance (avg)	36.5	35.75	34	32.6	32.5	32	
Load performance (avg)	59.5	54	54	52.25	52	49.75	
Noise Performance (By audibility)	Dull Hum	Silent	Silent	Silent	Silent	Silent	
Number of steps taken to install	5	8	5	8	8	8	

their combined width to a single and regular 120mm fan, their width would only be a few millimetres more. This space saving idea from Cooler Master helps the Hyper 412 Slim save space around the CPU socket. It gives some much needed space to work around the socket area and also the RAM slots making the Hyper 412 Slim a nice option to consider for Intel socket LGA 2011, since LGA 2011 has RAM slots on either side of the CPU socket.

We also received another twin tower design based cooler

from Deepcool, christened Deepcool NEPTWIN. The NEPTWIN was the biggest air cooler in this test, in the sense that it had two huge heatsinks connected by six heatpipes spread out in a linear fashion. Add to that two 120mm fans and you will know how big and heavy it is. It's size is the reason why it was tough to install and we literally had to lay down the test cabinet horizontally to install it and it required two pair of hands to mount properly.

The NEPTWIN is also big in terms of features, as it not

only has a mirror finished base but Deepcool has included a small fan hub along with the NEPTWIN which allows to connect four fans on it and power them. They've also included a 3M sticker and a cable tie which help in sticking or tying the fan hub inside your case where you desire. Performance wise, the NEPTWIN lies in the middle of the pack and looking at its price, it can be safe to say that there are better options to consider, some of which won't block the space all around the CPU socket and also your

motherboard's RAM slots are an ideal choice.

We also received the Cooler Master TPC 612 and the Cooler Master TPC 812 for the test. The TPC 612 and TPC 812, both use vertical vapor chambers along with the usual heat pipes to dissipate heat through their heatsinks. Designed similarly, with a nice touch of heat pipe caps on the top, the TPC 612 is slimmer and lighter than the TPC 812. As TPC 612 is slimmer, it uses four heat pipes instead of the six heat pipes seen on the TPC 812.